



PERMAFROST Digital Twin

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Presentation Overview

Problem & Context

- Background & Motivation
- Project Description

Proposed Solution

- Dataset Description
- Permafrost Modelling
- Digital Twin Architecture
- Visualization

Conclusion

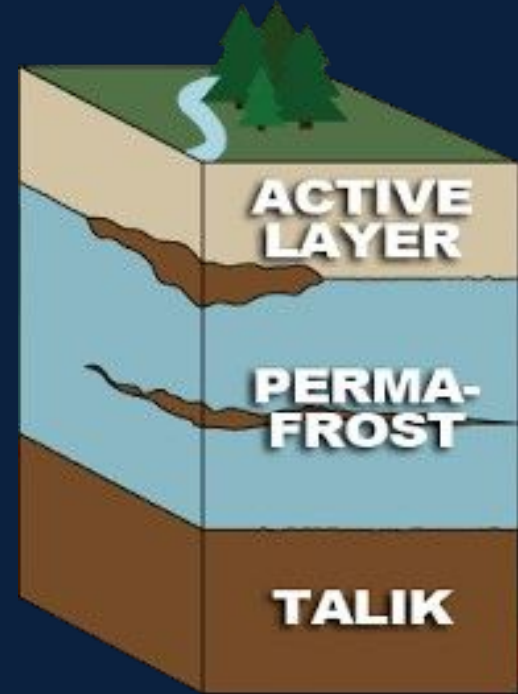
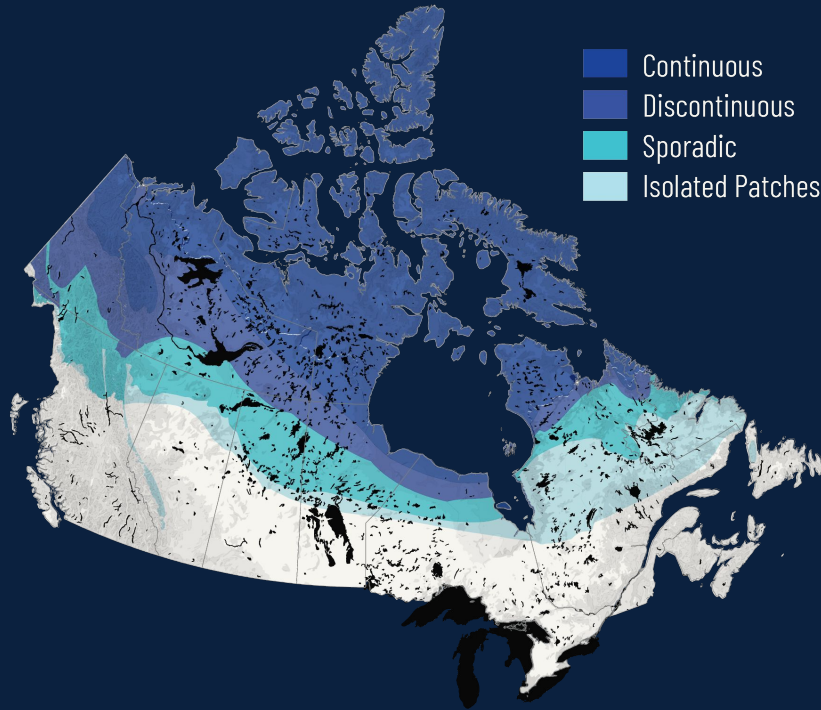
- Summary
- Future Work
- Challenges
- Q & A



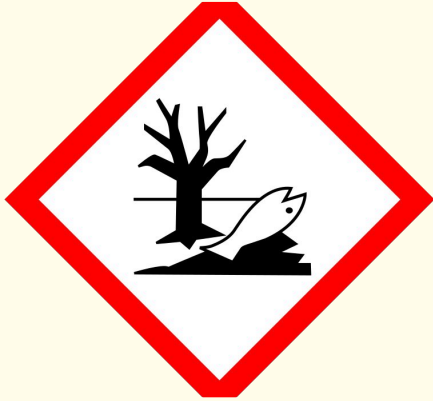
Background & Motivation

01

What is permafrost ?

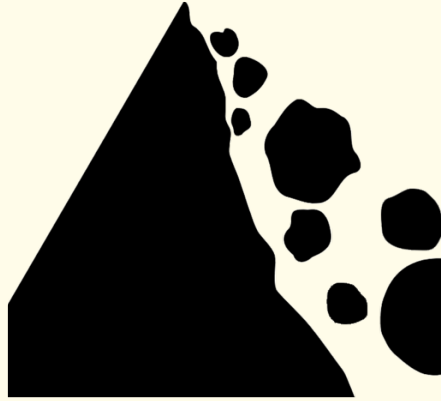


Risks of Permafrost Degradation



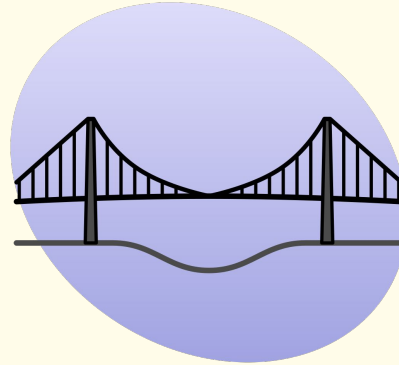
Ecosystem Damage

Disappearing lakes
Impact on wildlife and vegetation



Landslides

Banks Island in Canada's Arctic, 60x increase in landslides from 1984 to 2015



Infrastructure Damage

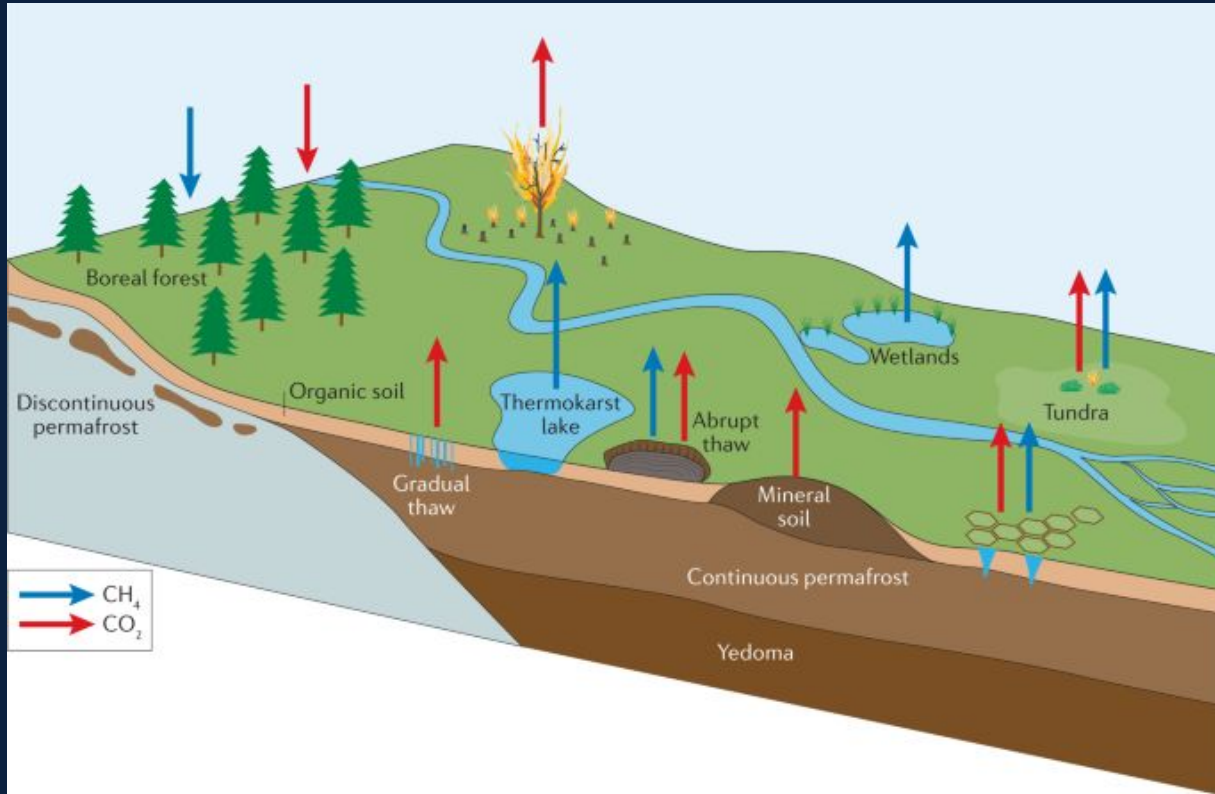
Threatens 30–50% of critical northern infrastructure by 2050



Acceleration of global warming

Accelerate future climate change by 10–20%

Permafrost Carbon Emission



Research Problem

- Importance of permafrost monitoring
- Difficulty of permafrost observation
- Borehole are expensive
~\$50,000–\$150,000+

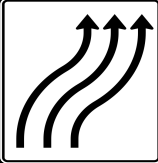
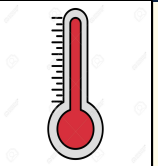
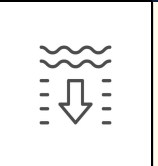
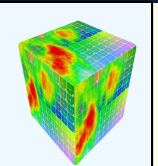
Project Description

03

Goal:
Develop a DT to predict,
monitor and visualize
temperature variations in
permafrost



Milestones

Modelling		Model heat transfer inside permafrost
Measurements		Use real temperature measurements at surface level
Estimations		Compute temperature values in depth using the model
Visualization & Monitoring		Visualize and monitor permafrost evolution

Dataset Description

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Dataset Description



Boreholes - Permafrost Temperatures

Draw selection zone Reset

Elevation Range (m) -20 5068

Depth Range (m) 0 1028

Search filters:

- cd
- Sites Name...
- Country Name...
- GTN-P Code...
- Vegetation Type -
- Permafrost Zone -
- Responsible Person -
- Variable -

Name	Site	Country	Gtn-P	Vegetation	Permafrost	Elevation	Depth	Data	Select
CD100	Mackenzie Delta	Canada	CA 129	Forest Tundra	Discontinuous	2.00	20.00	Yes	☐
CD101	Mackenzie Delta	Canada	CA 130	Forest Tundra	Discontinuous	2.00	18.00	Yes	☐
CD102	Mackenzie Delta	Canada	CA 131	Forest Tundra	Discontinuous	2.00	20.00	Yes	☐
CD103A	Mackenzie Delta	Canada	CA 132	Forest Tundra	Discontinuous	2.00	19.00	Yes	☐
CD104	Mackenzie Delta	Canada	CA 133	Forest Tundra	Discontinuous	2.00	20.00	Yes	☐
CD105	Mackenzie Delta	Canada	CA 134	Forest Tundra	Discontinuous	2.00	20.00	Yes	☐

<http://gtnpdatabase.org/boreholes>

Permafrost Modelling

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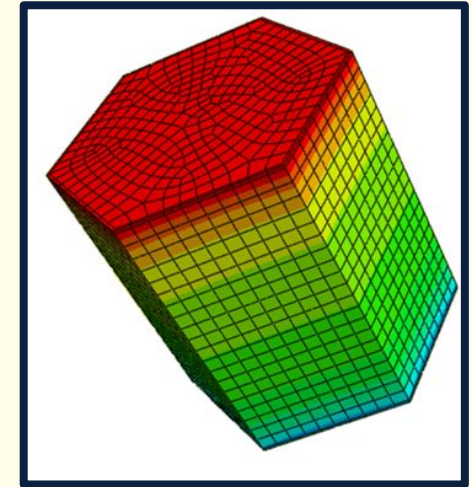
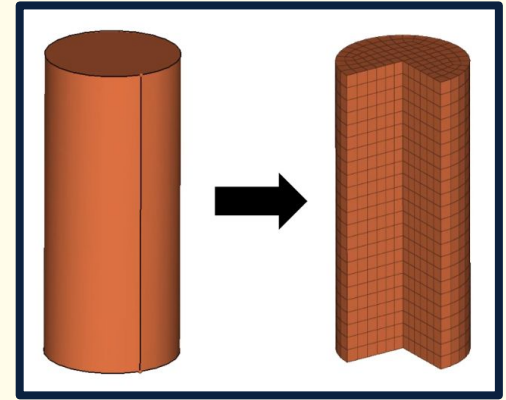
Heat Transfer Equation with Phase Change

$$(\rho c_p)_{\text{eff}} \frac{\partial T}{\partial t} + \nabla \cdot (\lambda_{\text{eff}} \nabla T) - L_f \rho_i \frac{\partial \theta_i}{\partial t} = 0$$

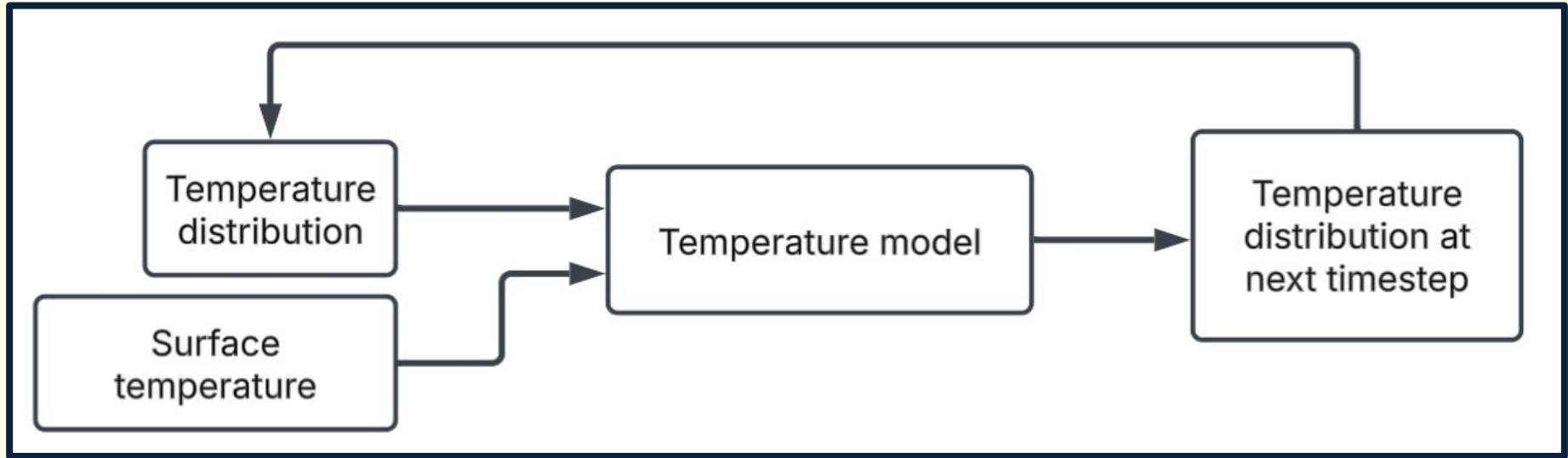
- Physical properties (Heat capacity, Thermal Conductivity...)
- Initial conditions (Temperature distribution at $t = 0$)
- Boundary conditions (Surface Temperature)

Finite Element Modelling

1. Spatial discretization
2. Temporal discretization
3. Equation solving for each element

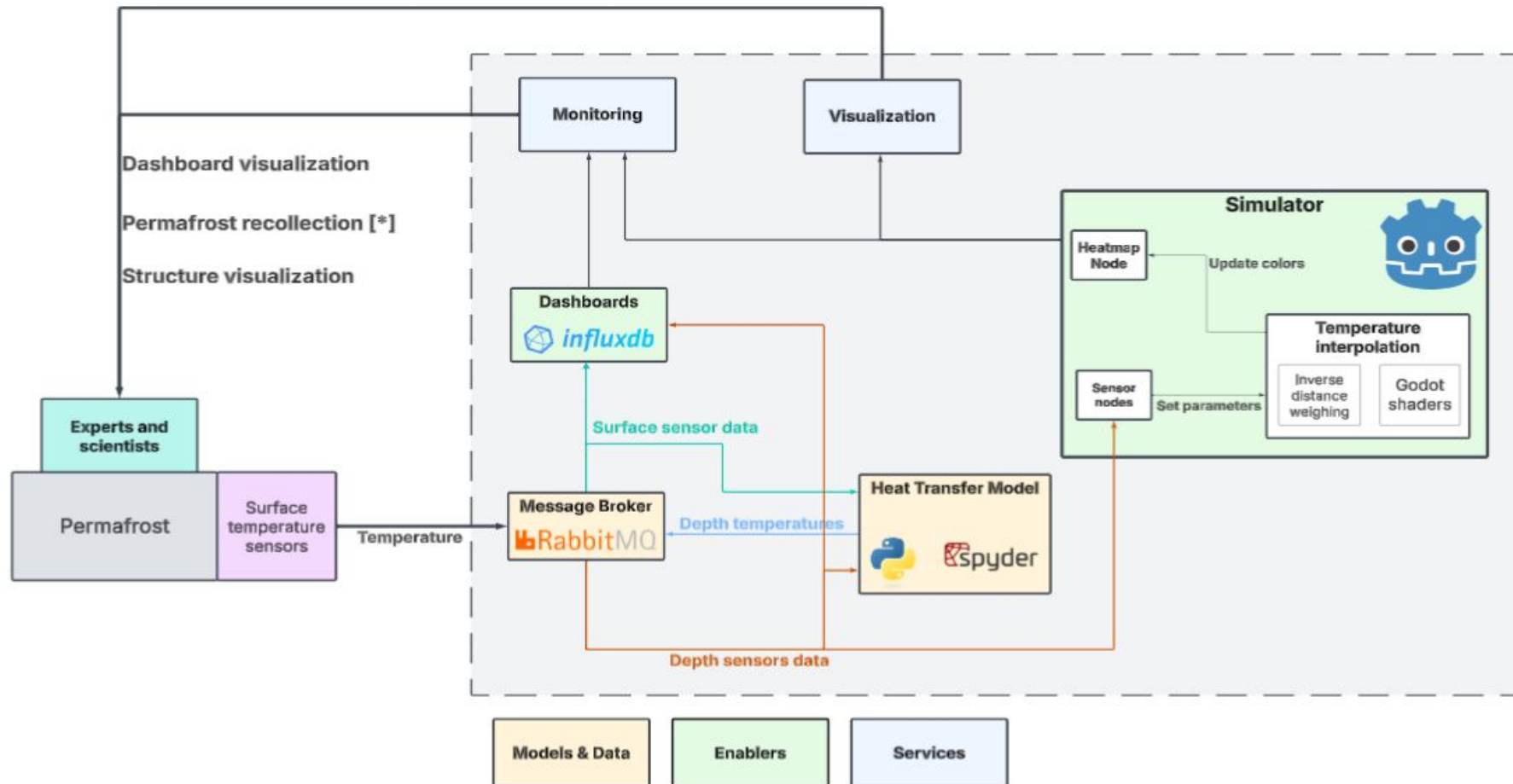


Model Integration



Digital Twin Architecture

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Visualization

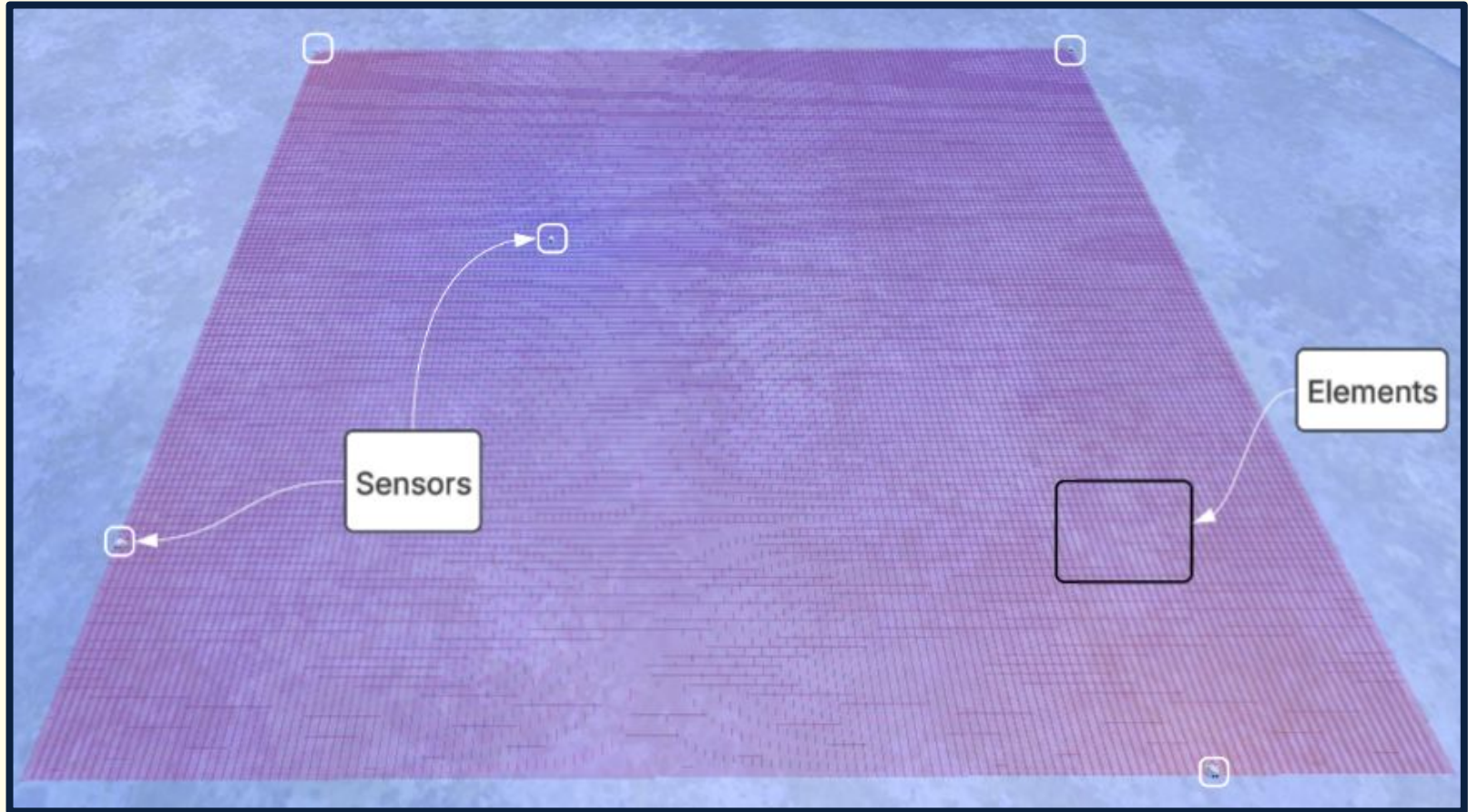
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Version 1: Inspired from finite element modelling

1. Place sensors
2. Discretize space
3. Interpolate
 - Too many generated meshes
 - Interpolation is too slow



Version 1

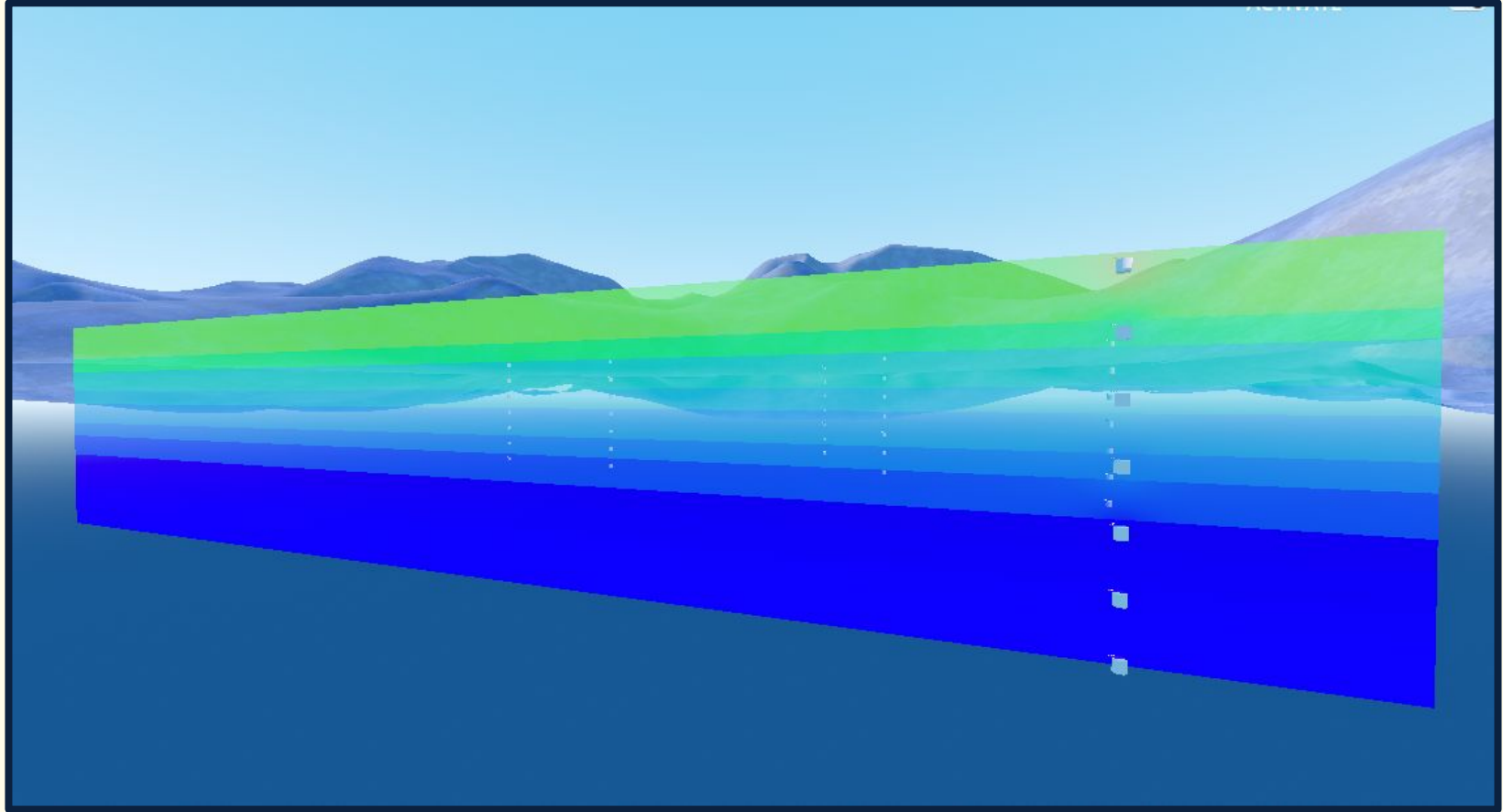


Version 2: Using shaders

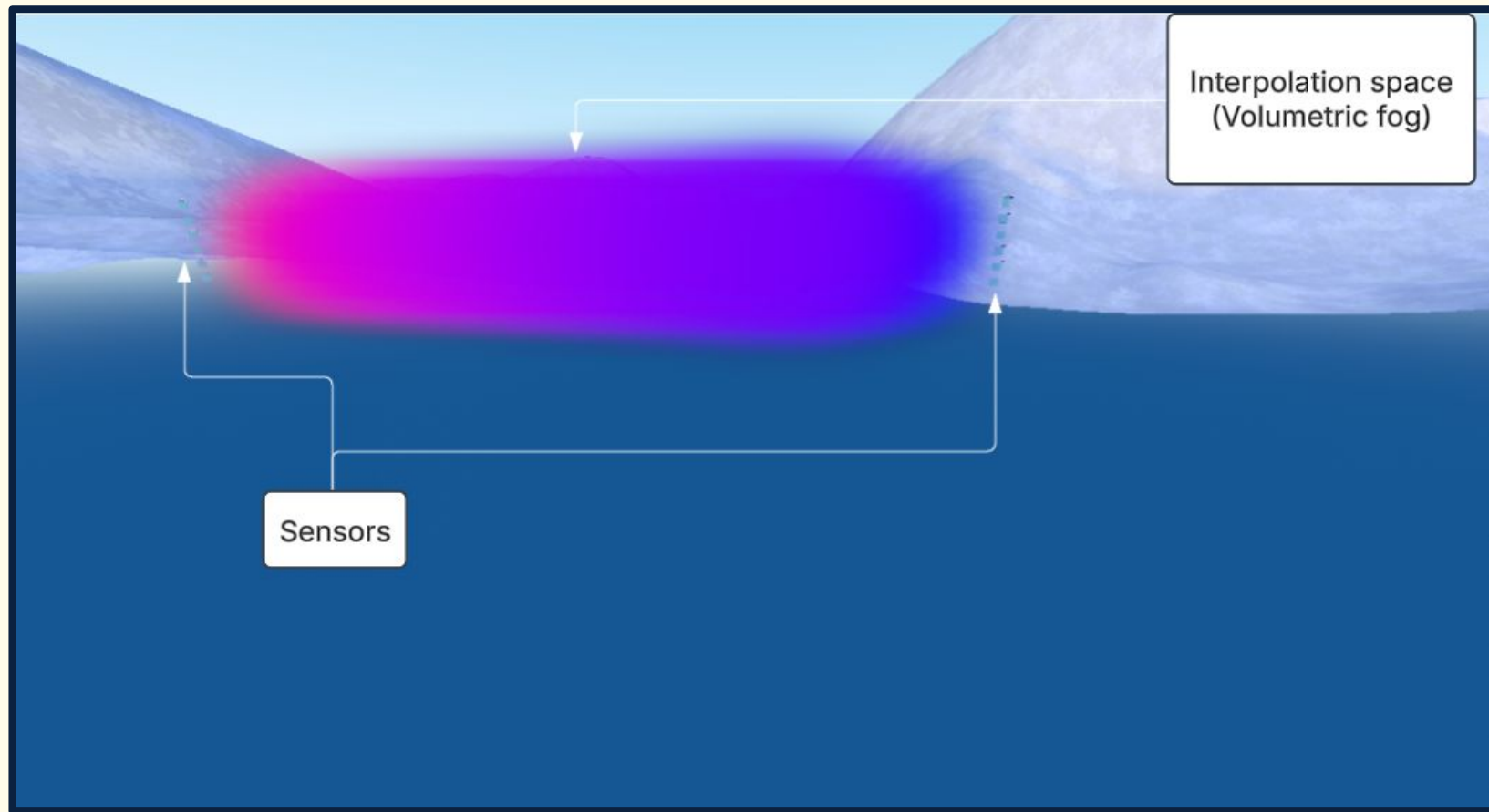
- Use only one mesh
- Interpolate in shader code (GPU)
- Cannot visualize the internal structure of the permafrost



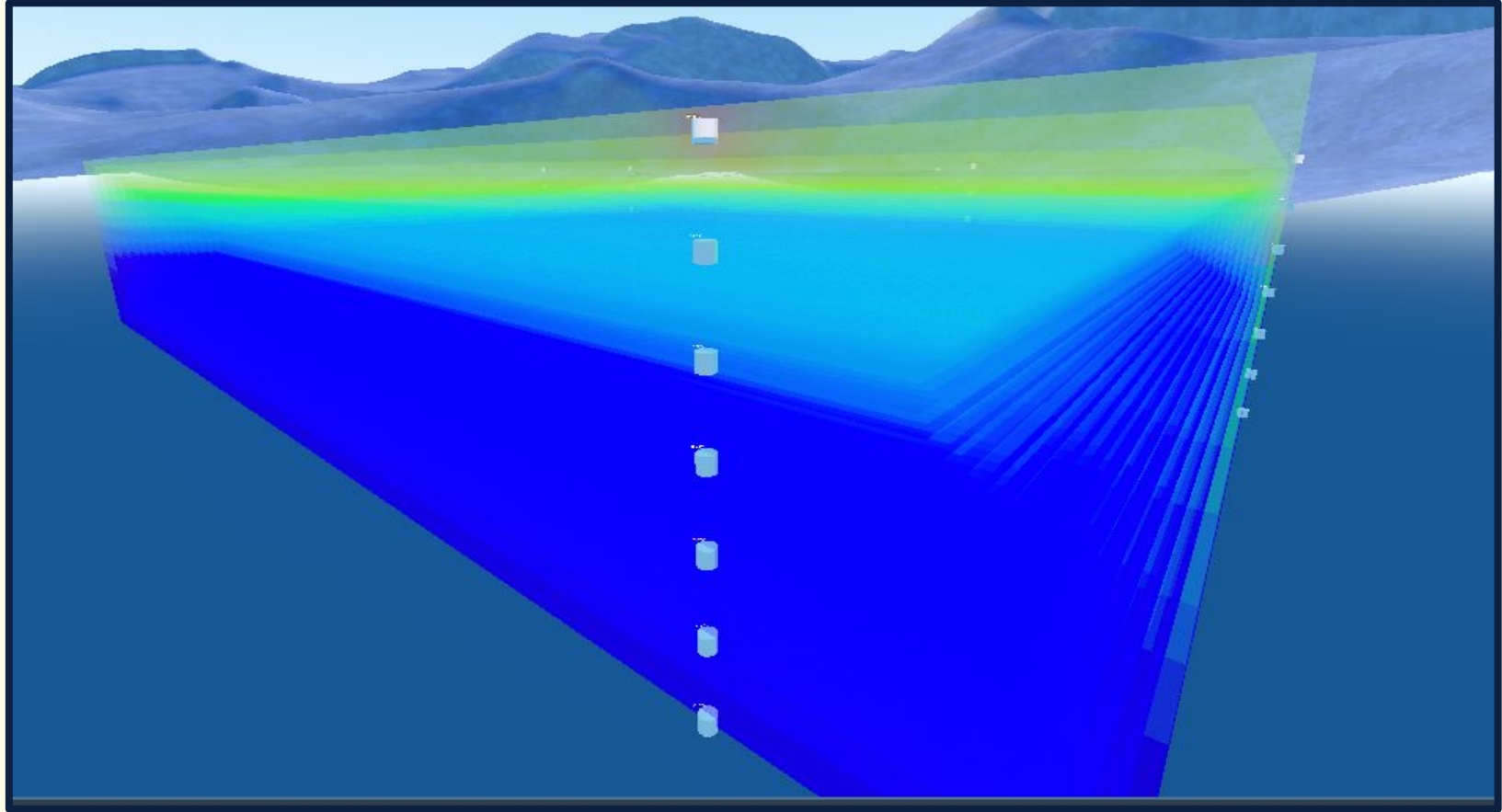
Version 2



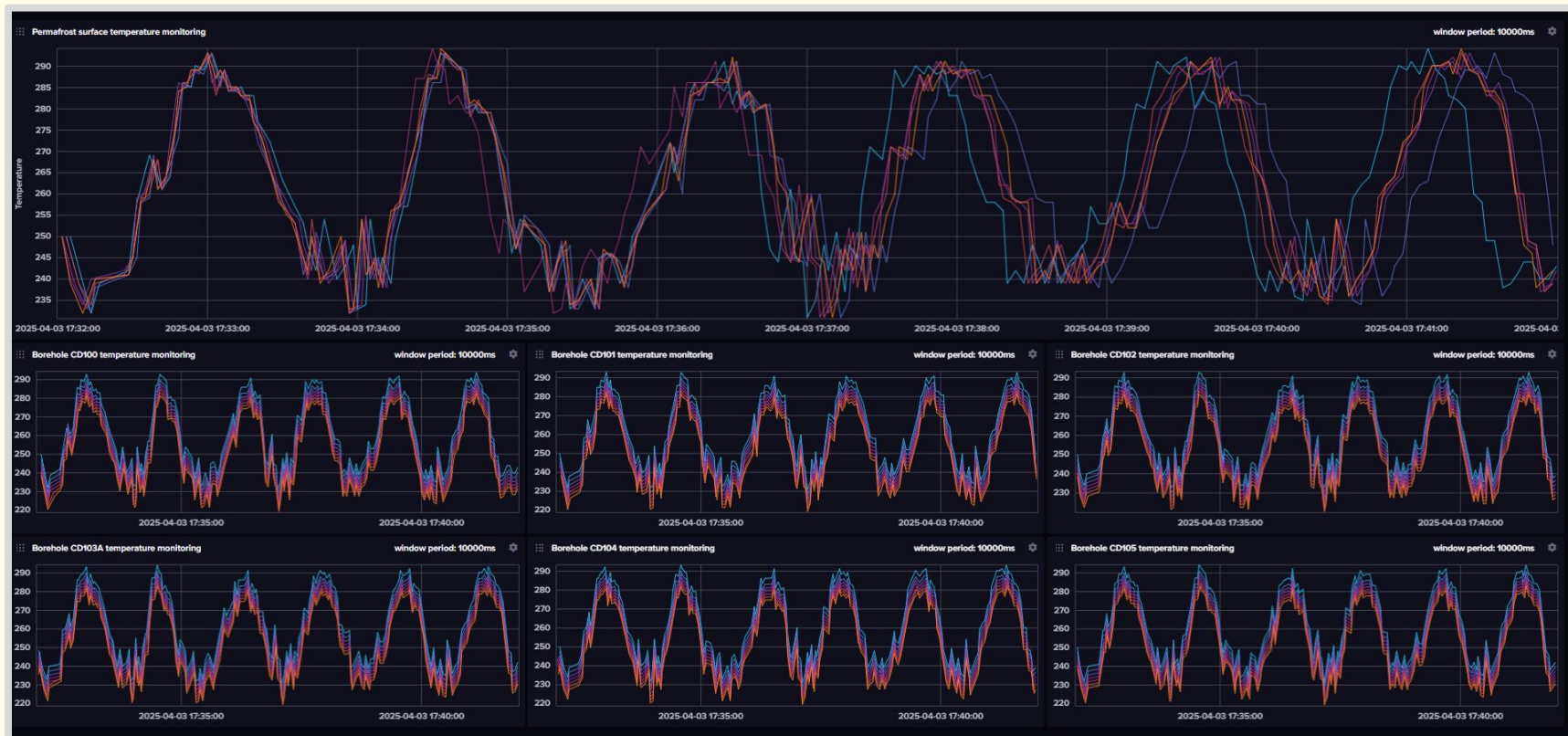
Version 3: Using volumetric fog shaders



Final version

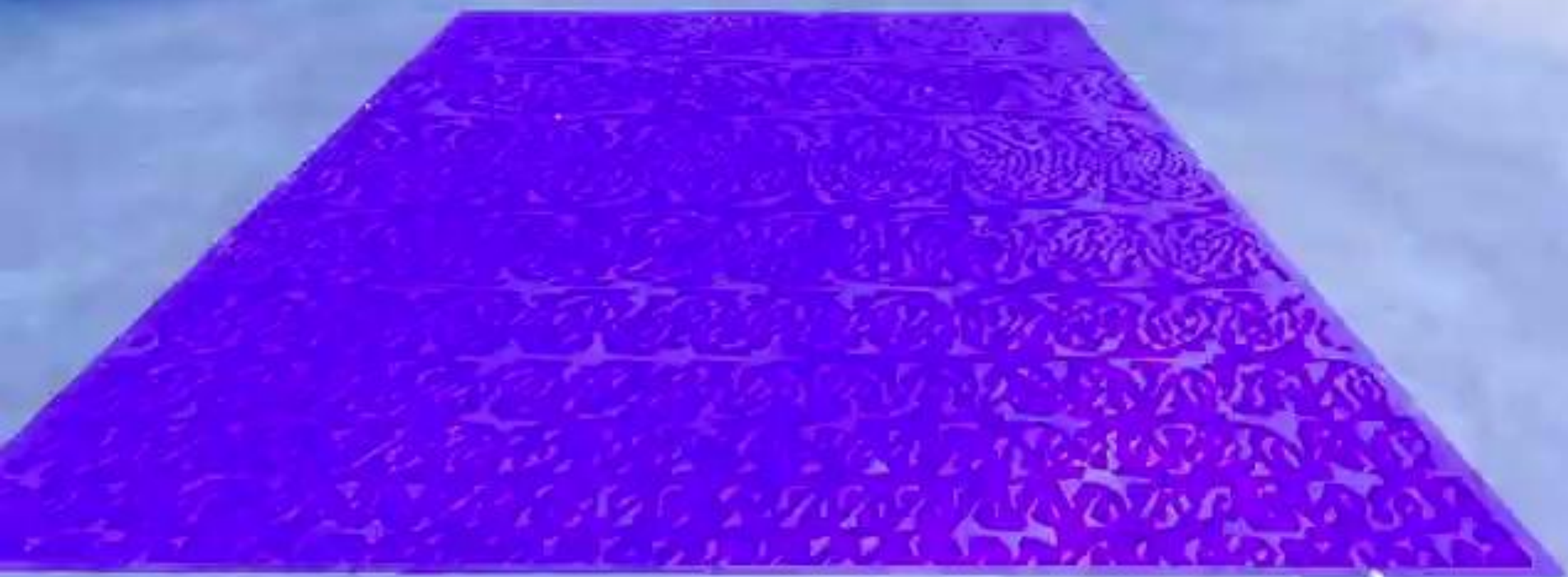


InfluxDB Monitoring



ACTIVATE

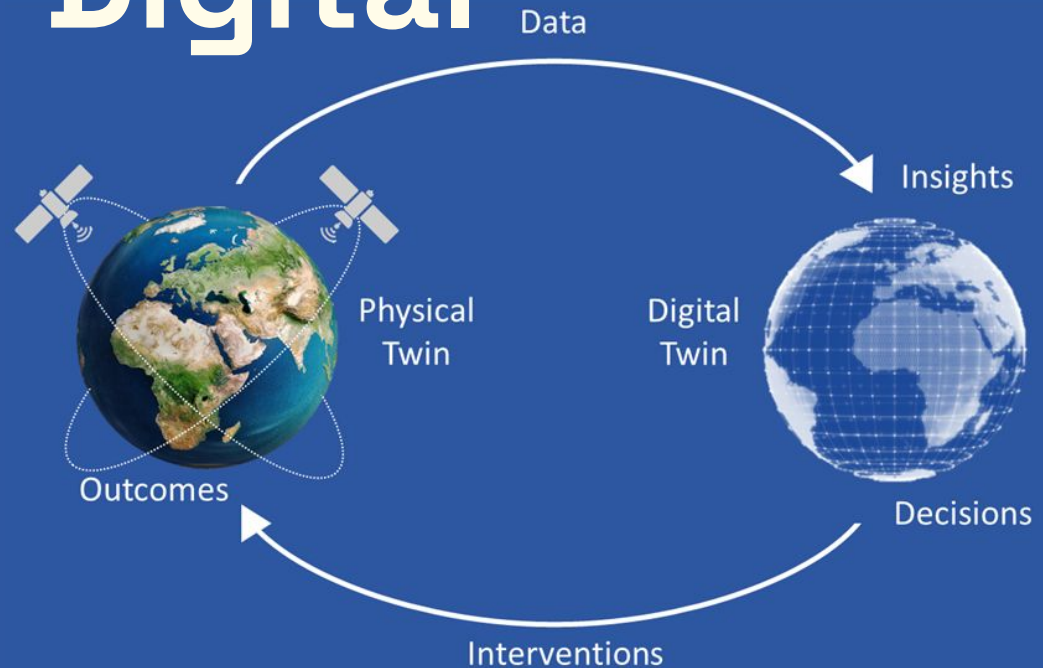




Summary

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Digital Twin, Digital Shadow, or Digital Model

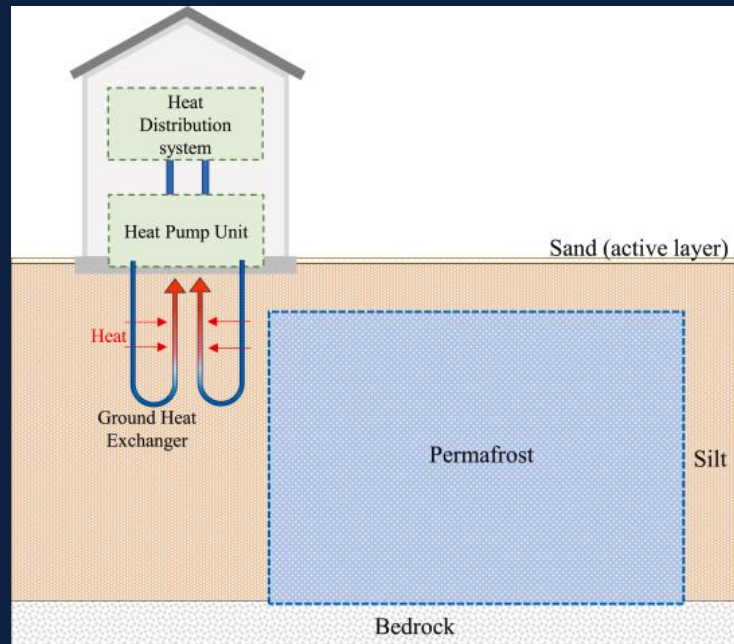


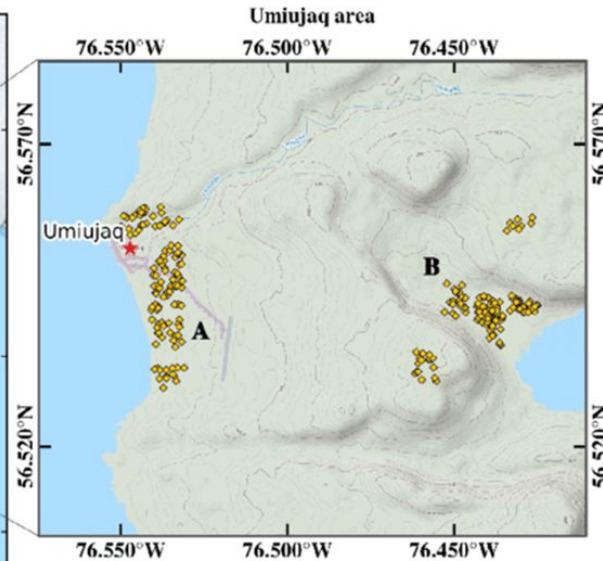
Future Work

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Ground Heat Source Pump (GHSP)

- Transfer heat to or from the ground
- Uses subsurface temperature to heat/cool buildings
- Leads to permafrost degradation
- DT can :
 - Simulate thermal impact on frozen ground
 - Predict thaw risk around GSHP installations
 - Optimize energy use





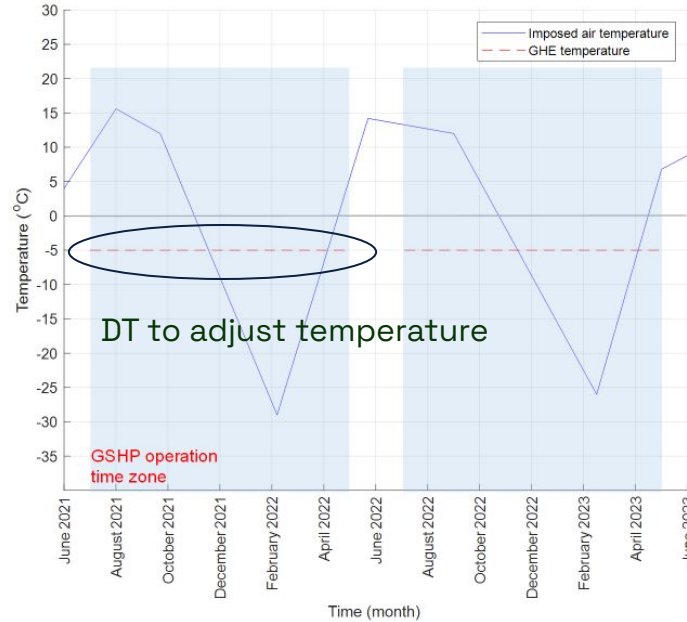
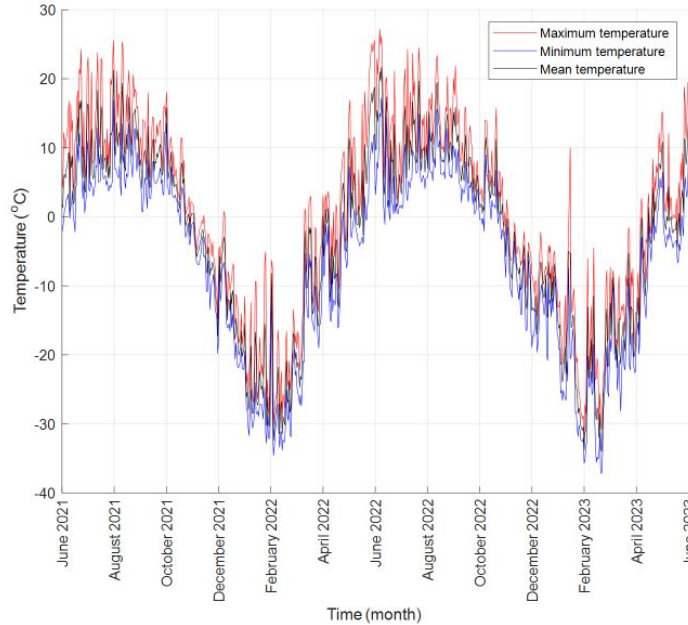
Legend

- ★ Umiujaq
- ◆ Shrub sampling plots

- A) Coastal area
- B) Tasiapik valley



Application to geo-engineering projects in cold regions

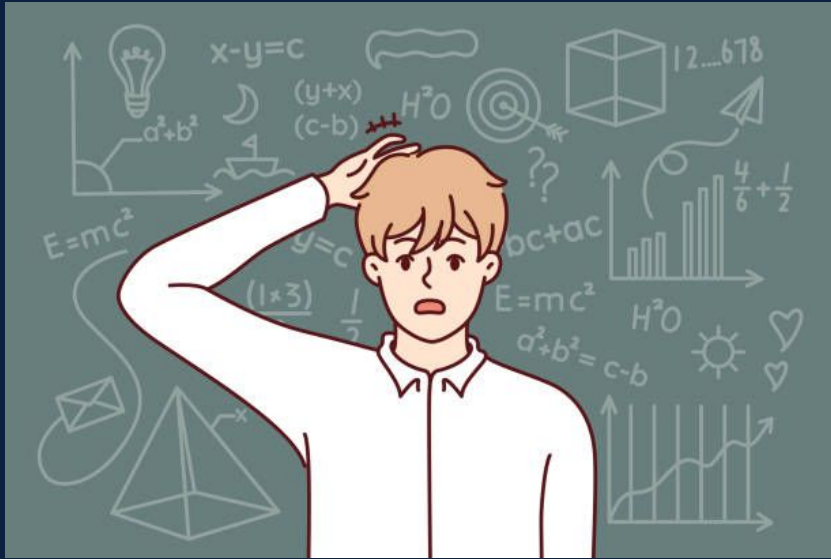


(a) Measured air and ground surface temperatures in **Umiujaq** from NRC and (b) temperature variations of air and surface as well as GHE in simulation.

Challenges

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Challenges



Complex Formulas



Confusing Communications



Thank you!
Questions?

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